For Examiner's Use

3 **(a)** State the principle of superposition.

……

……

…… [2]

(b) A ship at X is equidistant from two shore-based radio transmitters P and Q. The stations are 60 km apart, and each behaves as a point source of electromagnetic waves which radiate circular wavefronts. Both transmitters operate with a frequency of 1.0 MHz and radiate signals of equal amplitude.

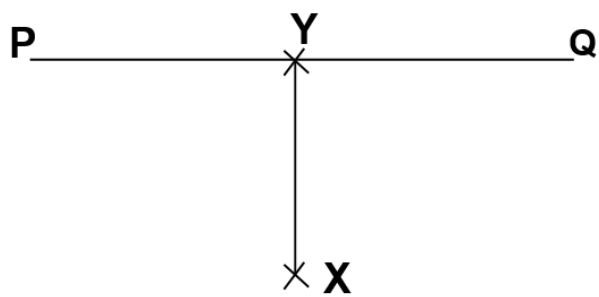

Fig. 3.1

(i) When the ship is at position **X** as shown in Fig. 3.1, the ship detects zero signal amplitude. State what can be deduced about the signals from P and Q.

……

…… [1]

(ii) The ship moves in a straight line from **X** to **Y**. Explain why, throughout the journey, the amplitude of the signal detected by the ship is zero.

……

……

……

…… [2]

(iii) The ship moves in the direction **YQ** until the signal detected has an amplitude twice that from either transmitter alone. Determine the distance that the ship has moved. Explain your answer.

For Examiner's Use

distance = …… m [3]

(iv) When the ship sails from **Y** to the harbour alongside transmitter **Q** the detected signal rises and falls in amplitude. Calculate how many dips in intensity will be passed.

number = …… [2]